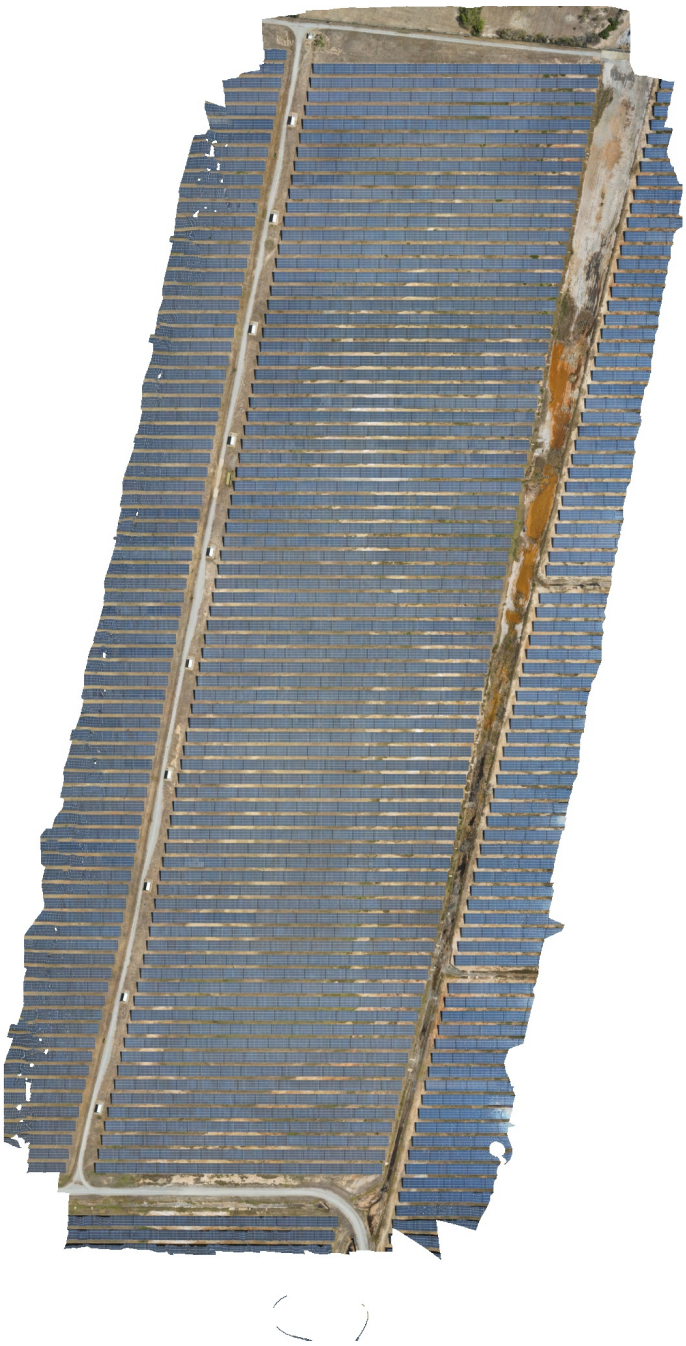
# Agisoft Metashape

**Processing Report**

**08 September 2026**



# Survey Data

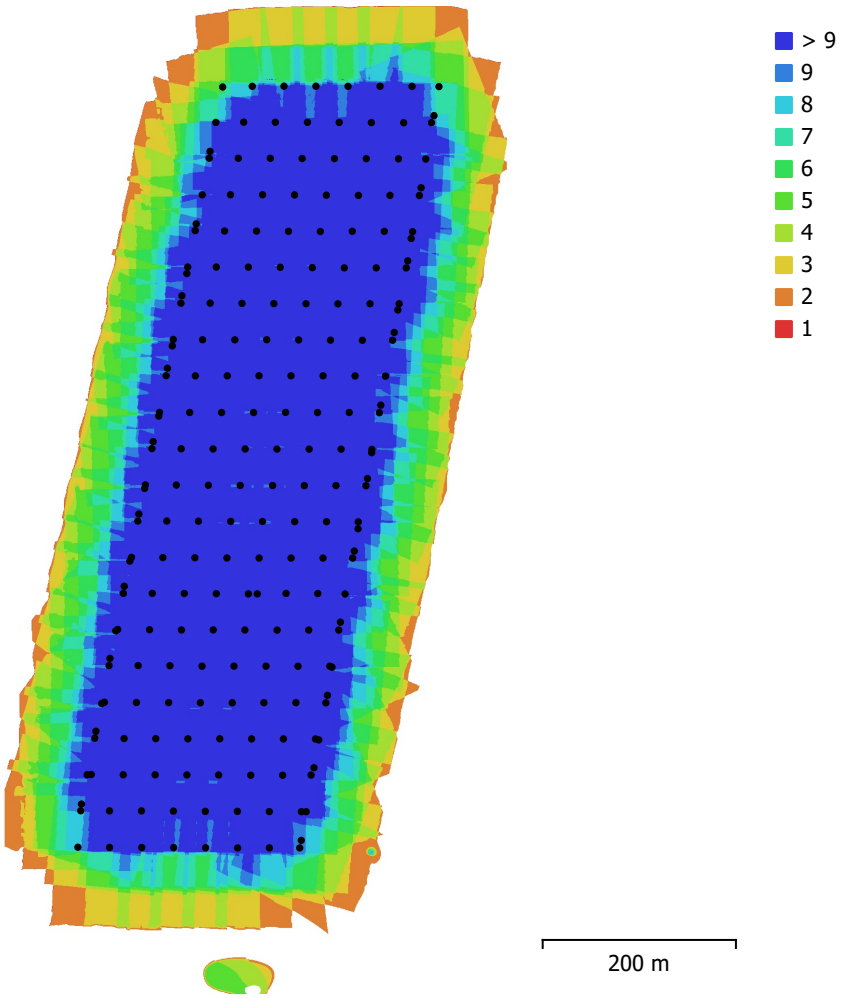



Fig. 1. Camera locations and image overlap.

| | | | |
|---|---|---|---|
| Number of images: | 213 | Camera stations: | 213 |
| Flying altitude: | 109 m | Tie points: | 132,402 |
| Ground resolution: | 1.86 cm/pix | Projections: | 409,808 |
| Coverage area: | 0.358 km² | Reprojection error: | 0.733 pix |

| Camera Model | Resolution | Focal Length | Pixel Size | Precalibrated |
|---|---|---|---|---|
| MAVIC2-ENTERPRISE-A... | 8000 x 6000 | 4.5 mm | 0.811 x 0.811 µm | No |

Table 1. Cameras.

# Camera Calibration

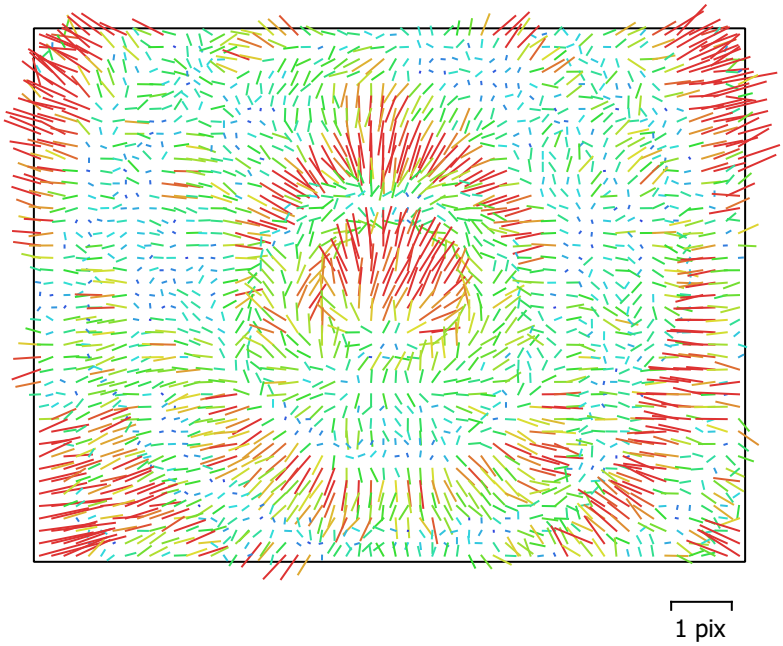



Fig. 2. Image residuals for MAVIC2-ENTERPRISE-ADVANCED (4.5mm).

**MAVIC2-ENTERPRISE-ADVANCED (4.5mm)**

213 images

| Type | Resolution | Focal Length | Pixel Size |
|---|---|---|---|
| **Frame** | **8000 x 6000** | **4.5 mm** | **0.811 x 0.811 µm** |

| | Value | Error | Cx | Cy | B1 | B2 | K1 | K2 | K3 | P1 | P2 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **F** | **5547** | | | | | | | | | | |
| **Cx** | **-19.5739** | 0.31 | 1.00 | 0.28 | 0.26 | -0.31 | 0.13 | 0.01 | -0.02 | -0.18 | 0.05 |
| **Cy** | **43.1931** | 0.19 | | 1.00 | 0.25 | 0.05 | -0.04 | 0.00 | 0.01 | -0.05 | -0.00 |
| **B1** | **-4.84732** | 0.018 | | | 1.00 | -0.06 | -0.02 | -0.04 | 0.04 | -0.05 | -0.12 |
| **B2** | **1.52221** | 0.017 | | | | 1.00 | -0.07 | -0.00 | 0.01 | 0.07 | -0.04 |
| **K1** | **-0.00358979** | 8.6e-05 | | | | | 1.00 | -0.46 | 0.37 | -0.22 | 0.43 |
| **K2** | **-0.018385** | 0.00011 | | | | | | 1.00 | -0.98 | 0.01 | -0.02 |
| **K3** | **0.00477468** | 9.4e-05 | | | | | | | 1.00 | -0.01 | -0.01 |
| **P1** | **-7.10149e-05** | 2.8e-06 | | | | | | | | 1.00 | -0.07 |
| **P2** | **-0.000306142** | 2.6e-06 | | | | | | | | | 1.00 |

Table 2. Calibration coefficients and correlation matrix.

# Camera Locations

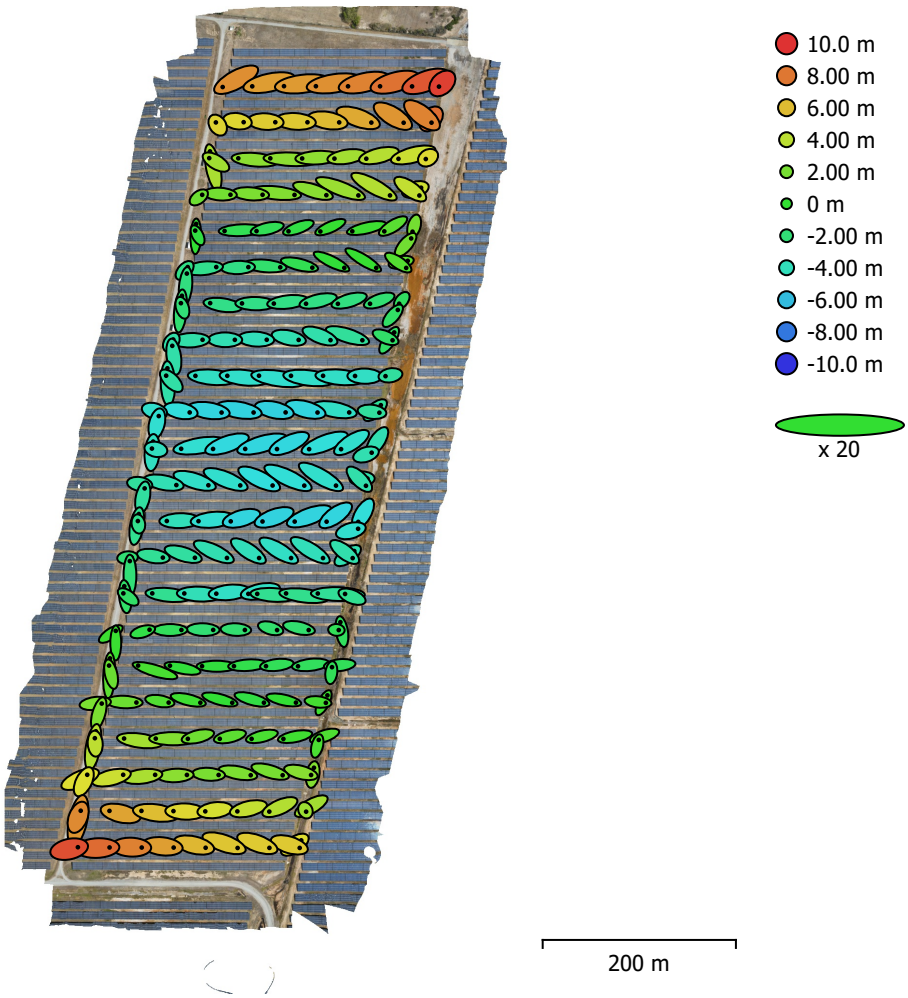



Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

| X error (m) | Y error (m) | Z error (m) | XY error (m) | Total error (m) |
|---|---|---|---|---|
| 1.13469 | 0.546861 | 3.95253 | 1.2596 | 4.14838 |

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

# Digital Elevation Model

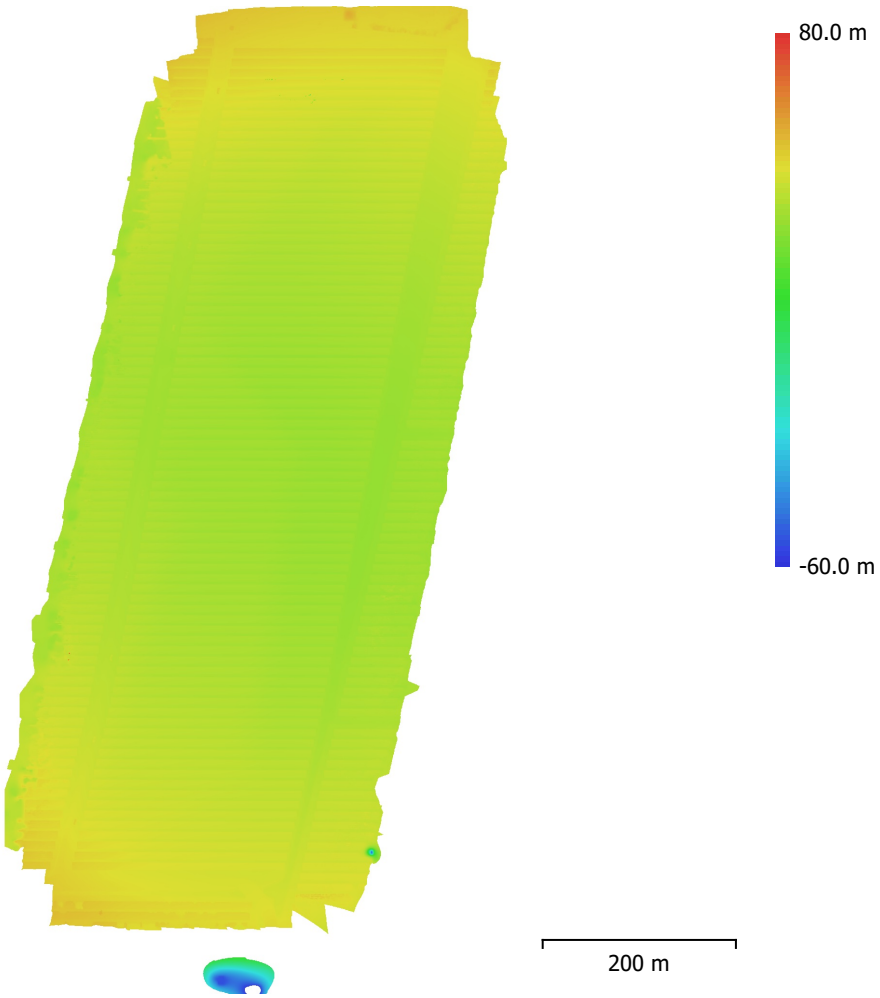



Fig. 4. Reconstructed digital elevation model.

| | |
|---|---|
| Resolution: | 3.92 cm/pix |
| Point density: | 652 points/m² |

# Processing Parameters

| | |
|---|---|
| **General** | |
| Images | 213 |
| Aligned images | 213 |
| Coordinate system | WGS 84 (EPSG::4326) |
| Rotation angles | Yaw, Pitch, Roll |
| **Tie Points** | |
| Points | 132,402 of 276,968 |
| RMS reprojection error | 0.242049 (0.732516 pix) |
| Max reprojection error | 2.39965 (15.1268 pix) |
| Mean key point size | 3.18971 pix |
| Point colors | 3 bands, uint8 |
| Key points | 700.41 MB |
| Average tie point multiplicity | 2.98185 |
| **Alignment parameters** | |
| Accuracy | High |
| Generic preselection | Yes |
| Reference preselection | Source |
| Key point limit | 40,000 |
| Key point limit per Mpx | 1,000 |
| Tie point limit | 4,000 |
| Exclude stationary tie points | Yes |
| Guided image matching | No |
| Adaptive camera model fitting | Yes |
| Matching time | 6 minutes 10 seconds |
| Matching memory usage | 336.95 MB |
| Alignment time | 1 minutes 46 seconds |
| Alignment memory usage | 321.55 MB |
| **Optimization parameters** | |
| Parameters | f, cx, cy, k1, k2, p1, p2 |
| Adaptive camera model fitting | Yes |
| Exclude corners | No |
| Optimization time | 2 seconds |
| Date created | 2026:09:05 08:44:39 |
| Software version | 2.3.1.22580 |
| File size | 720.02 MB |
| **Depth Maps** | |
| Count | 213 |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 46 minutes 17 seconds |
| Memory usage | 8.04 GB |
| Date created | 2026:09:05 09:44:26 |
| Software version | 2.3.1.22580 |
| File size | 3.62 GB |
| **Point Cloud** | |
| Points | 269,633,261 |
| Coordinate precision | 9.53 m |
| **Point attributes** | |

| | |
|---|---|
| Color | 3 bands, uint8 |
| Normal | |
| **Point classes** | |
| Created (never classified) | 269,633,261 |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 46 minutes 17 seconds |
| Memory usage | 8.04 GB |
| **Point cloud generation parameters** | |
| Source data | Depth maps |
| Processing time | 55 minutes 47 seconds |
| Memory usage | 10.19 GB |
| Date created | 2026:09:05 10:40:20 |
| Software version | 2.3.1.22580 |
| File size | 3.50 GB |
| **DEM** | |
| Size | 13,254 x 26,085 |
| Resolution | 3.92 cm/pix |
| Coordinate system | WGS 84 (EPSG::4326) |
| **Depth maps generation parameters** | |
| Quality | High |
| Filtering mode | Aggressive |
| Max neighbors | 16 |
| Processing time | 46 minutes 17 seconds |
| Memory usage | 8.04 GB |
| **Reconstruction parameters** | |
| Source data | Depth maps |
| Interpolation | Enabled |
| Processing time | 1 hours 23 minutes |
| Memory usage | 7.79 GB |
| Date created | 2026:09:05 22:56:43 |
| Software version | 2.3.1.22580 |
| File size | 959.15 MB |
| **Orthomosaic** | |
| Size | 26,508 x 52,170 |
| Resolution | 1.96 cm/pix |
| Coordinate system | WGS 84 (EPSG::4326) |
| Colors | 3 bands, uint8 |
| Orthophotos | 3.00 GB |
| **Reconstruction parameters** | |
| Blending mode | Mosaic |
| Surface | DEM |
| Enable hole filling | Yes |
| Enable ghosting filter | No |
| Processing time | 12 minutes 58 seconds |
| Memory usage | 4.55 GB |
| Date created | 2026:09:05 23:02:16 |
| Software version | 2.3.1.22580 |
| File size | 3.89 GB |
| **System** | |
| Software name | Agisoft Metashape Professional |
| Software version | 2.3.0 build 21954 |
| OS | Windows 64 bit |
| RAM | 15.78 GB |

| | |
|---|---|
| CPU | 11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz |
| GPU(s) | NVIDIA GeForce RTX 3050 Laptop GPU |